

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}.\frac{1}{945}.$$

[illegible]

Simplify deltas

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot-\frac{1}{945}.$$

[illegible]

Simplify R

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot-\frac{1}{945}.$$

[illegible]

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945} \cdot \left[\begin{array}{l} +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \end{array} \right] \quad (4a)$$

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Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon} = R_z^{-13} \cdot -\frac{1}{945} \cdot \left[\begin{array}{l} +105 \cdot R_z^0 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \end{array} \right] \quad (5a)$$

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Reduce Indices

[illegible]

Add up

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot -\frac{1}{945} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (7)$$

Combining everything

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = \begin{bmatrix} +5/9 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +10/9 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +5/3 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -1/3 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ -4/3 \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ +4/21 \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (8)$$